

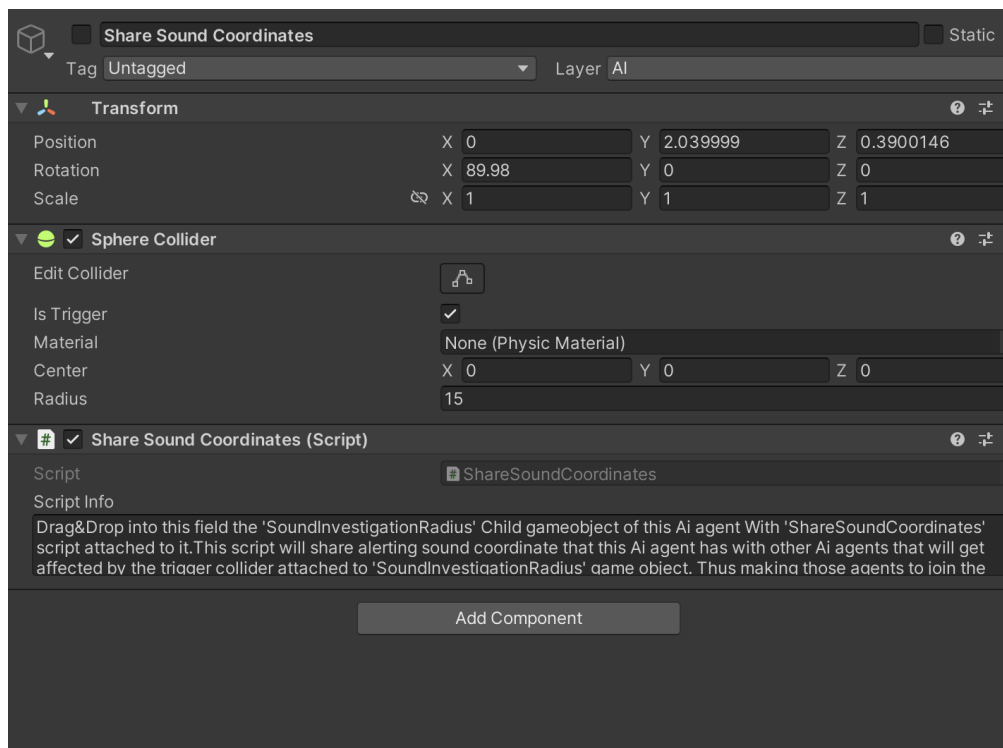
Sound Alert Behaviour Part-5

Introduction

In this tutorial you will learn how AI agent share sound coordinate with his friendly AI agent and how to customise AI agent approach towards the sound coordinate.

Share Sound Coordinate

This script will share alerting sound coordinate that this AI agent has with other AI agents that will get affected by the trigger collider attached to 'SoundInvestigationRadius' game object. Thus making those agents to join the sound investigation. Those Agents that didn't hear the alerting sound but joined the investigation will also enable their own 'SoundInvestigationRadius' colliders to affect other unaware AI agents of this investigation. Such design allows for a domino effect where first alerted AI agent is alerting other nearby AI agents.

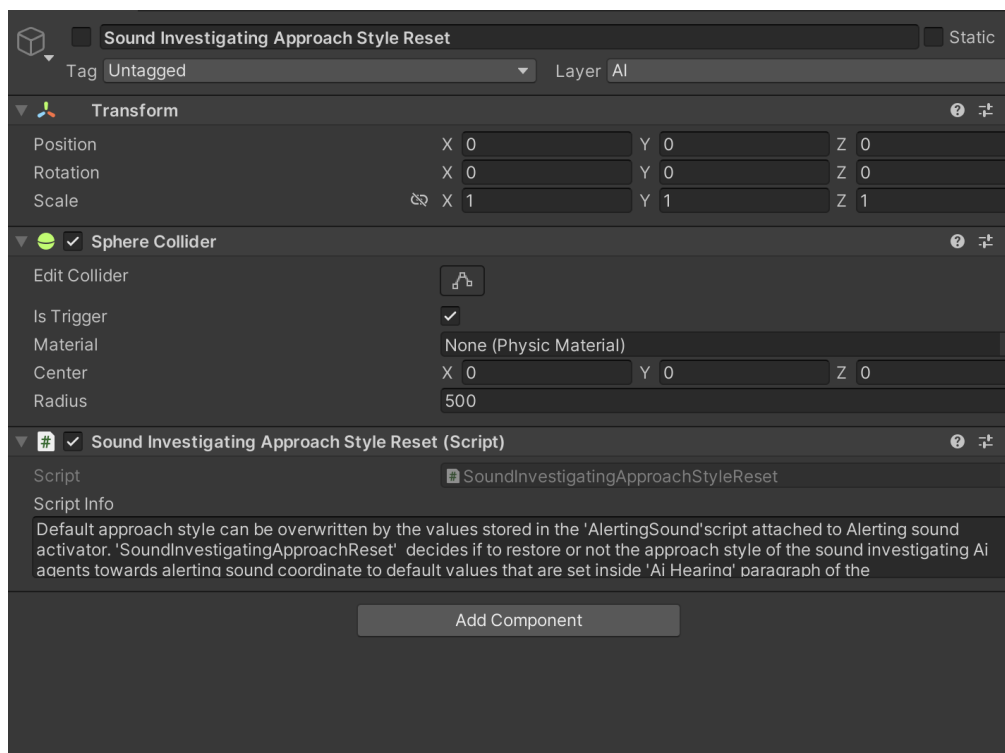


Sound Investigating Approach Style Reset

Default approach style can be overwritten by the values stored in the 'AlertingSound' script attached to Alerting sound activator.

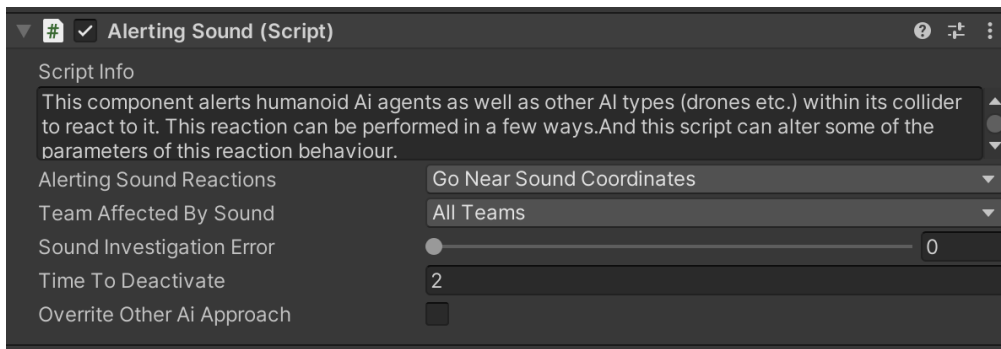
'SoundInvestigatingApproachReset' decides if to restore or not the approach style of the sound investigating Ai agents towards alerting sound coordinate to default values that are set inside 'Ai Hearing' paragraph of the 'CoreAiBehaviour'script.

'InvestigatingApproachReset' gets activated for a brief moment upon Ai agent's death and checks if there are alive friendly Ai agents in the combat state as well as approaching sound alert investigators within it's collider radius.If there are alive friendly Ai agents that are in combat state within this collider at the moment of this check then no approach style reset is applied to investigators inside this collider. If there are no alive friendly Ai agents in combat state at the moment of check then approach style reset will be made to investigators inside this collider.



Alerting Sound

This component alerts humanoid Ai agents as well as other AI types (drones etc.) within its collider to react to it. This reaction can be performed in a few ways.And this script can alter some of the parameters of this reaction behaviour.

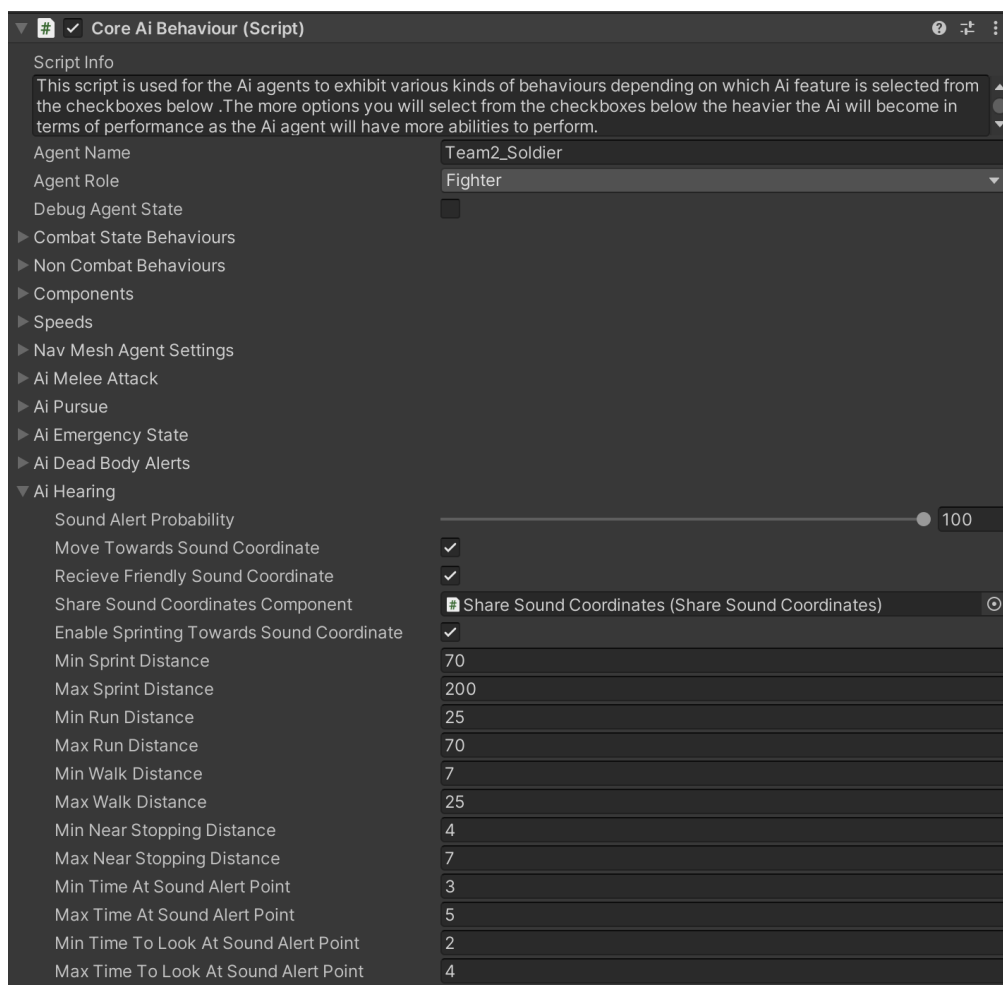


Fields	Info
AlertingSoundReactions	Choose one of the 2 behavioural responses of Ai agents affected by this sound alert. 'GoNearSoundCoordinates' to go towards the coordinate of this sound, or choose 'ForceEmergencyState' for Ai agent to go into emergency state.
TeamAffectedBySound	Choose the team which will be affected by this sound.
TeamName	Type the name of the team to be affected by this sound.
SoundInvestigationError	This field represents the sound location mistake the Ai agent will make. The greater the number, the further away the resulted investigation coordinate will get from the initial coordinate of the alerting sound.
TimeToDeactivate	Specify the time to deactivate this alerting sound game object.
OverrideOtherAiApproach	If checked, it will override the approach style of sound investigating Ai agents. If only one of those options is checked, then it will force itself on the way Ai agent will approach the sound and it will discard all the other approach styles. If two of those options are checked, then they will discard unchecked approach styles altogether. Quick Note - If all of checkboxes are checked, it will have the same effect as if none of them were checked, i.e., it will not take any effect over initial approach settings.
AllowSpringing	

	Allows only selected approach styles. If all or none of these options are selected, then default approach values will not be overridden.
AllowRunning	Allows only selected approach styles. If all or none of these options are selected, then default approach values will not be overridden.
AllowWalking	Allows only selected approach styles. If all or none of these options are selected, then default approach values will not be overridden.

AI Hearing

This paragraph contains fields related to AI agent reactions to alerting sounds.



Fields	Info
SoundAlertProbability	

	This slider sets the probability of this AI agent to react to sound alerts.
MoveTowardsSoundCoordinate	If checked, then Ai agent will move towards the alerting coordinate. Otherwise, he will look in the direction of that sound for a specified amount of time.
RecieveFriendlySoundCoordinate	If checked, then AI agent will join his friendly in alerting sound investigation if that friendly affects this agent with his sound alert radius. That secondary sound alert radius communicates the sound alert to all agents that got within this radius (even in cases when those agents are having zero probability to move out towards initial alerting sound that they heard themselves).
ShareSoundCoordinatesComponent	Reference to the component responsible for sharing sound coordinates.
EnableSprintingTowardsSoundCoordinate	Enable/disable sprinting towards sound coordinates.
MinSprintDistance	Minimal remaining distance to initial sound alert coordinate for the switch from walking or running to sprinting towards it.
MaxSprintDistance	Maximal remaining distance to initial sound alert coordinate for the switch from walking or running to sprinting towards it.
MinRunDistance	Minimal remaining distance to initial sound alert coordinate for the switch from walking or sprinting to running towards it.
MaxRunDistance	Maximal remaining distance to initial sound alert coordinate for the switch from walking or sprinting to running towards it.

MinWalkDistance	Minimal remaining distance to initial sound alert coordinate for the switch from running or sprinting to walking towards it.
MaxWalkDistance	Maximal remaining distance to initial sound alert coordinate for the switch from running or sprinting to walking towards it.
MinNearStoppingDistance	Minimum remaining distance to stop near the initial sound alert coordinate and to consider it as reached.
MaxNearStoppingDistance	Maximum remaining distance to stop near the initial sound alert coordinate and to consider it as reached.
MinTimeAtSoundAlertPoint	Set the minimum time the AI agent waits at the initial sound alert coordinate before returning to idle state.
MaxTimeAtSoundAlertPoint	Set the maximum time the AI agent waits at the initial sound alert coordinate before returning to idle state.
MinTimeToLookAtSoundAlertPoint	Set the minimum time the Stationary AI agent or agent with zero probability of moving towards initial sound alert coordinate, will look in the direction of the initial sound alert coordinate before returning to idle state.
MaxTimeToLookAtSoundAlertPoint	Set the maximum time the Stationary AI agent or agent with zero probability of moving towards initial sound alert coordinate, will look in the direction of the initial sound alert coordinate before returning to idle state.